FYUGP B.Sc. MATHEMATICS HONOURS

ABSTRACT ALGEBRA I (MAT5CJ302)

Semester V · Core in Major

MODULE 3: PERMUTATIONS, ORBITS, EVEN AND ODD PERMUTATIONS, COSETS AND LAGRANGE

Model questions on topics the past papers miss

Section A: 1 mark each, answer all

1 Is $(1,2)(2,3) = (2,3)(1,2)$ in S_3? Answer yes or no. **(1 mark)**

2 State whether true or false: if H is a subgroup of a finite group G, then the index $(G : H)$ equals $|G|/|H|$. **(1 mark)**

3 Find the orbit of 3 under the permutation $\sigma = (1,2)(3,4,5)(6)$ of $\{1,2,3,4,5,6\}$. **(1 mark)**

4 How many elements does A_3 have? **(1 mark)**

Section B: 2 marks each, answer all

5 Write the cycle $(1,2,3,4,5)$ as a product of transpositions. **(2 marks)**

6 Find all orbits of the permutation $\sigma = (1,4)(2,3,5)$ in S_5. **(2 marks)**

7 Define the left coset aH of a subgroup H of a group G, for $a \in G$. **(2 marks)**

8 Find the index of $\langle 4 \rangle$ in $\mathbb{Z}_{12}$. **(2 marks)**

9 Is the permutation $(1,2,3)(4,5)$ of S_5 even or odd? Justify your answer. **(2 marks)**

Section C: 4 marks each, answer any two

10 Let H be a subgroup of a group G. Show that the left cosets of H partition G. **(4 marks)**

11 Let H be a subgroup of a group G and $a \in G$. Show that the coset aH has the same number of elements as H. **(4 marks)**

12 Let $\phi : G \to G'$ be a group homomorphism with kernel K. Show that for $a, b \in G$, $a^{-1}b \in K$ if and only if $\phi(a) = \phi(b)$. **(4 marks)**

13 Let $\operatorname{sgn} : S_n \to \{1, -1\}$ be the sign of a permutation. Show that sgn is a group homomorphism. **(4 marks)**

14 Let σ be a cycle of length k in S_n. Show that σ is even if and only if k is odd. **(4 marks)**

Section D: 10 marks each, answer any one

15 Let G be a finite group and H a subgroup. Prove that the number of left cosets of H in G is $|G|/|H|$, and deduce that the order of every element of G divides $|G|$. **(10 marks)**

16 Let $\sigma \in S_n$ be written as a product of disjoint cycles. Show that σ is even if and only if it has an even number of cycles of even length. **(10 marks)**

MODULE 3: MODEL ANSWERS

Key steps for checking your work

Section A.

1. No. $(1,2)(2,3) = (1,2,3)$ and $(2,3)(1,2) = (1,3,2)$ (Table 4.15).
2. True. Each left coset has $|H|$ elements and the cosets partition G (Definition 10.10; Theorem 10.7).
3. $\{3,4,5\}$ (Section 8, orbits before Example 8.18; the cycle $(3,4,5)$ contains 3).
4. 3. $A_3 = \{\iota, (1,2,3), (1,3,2)\}$ (Theorem 8.24).

Section B.

5. $(1,2,3,4,5) = (1,5)(1,4)(1,3)(1,2)$, read right to left (8.14).
6. The orbits are $\{1,4\}$ and $\{2,3,5\}$ (orbits in Section 8 and Section 4).
7. $aH = \{ah \mid h \in H\}$ (Definition 10.2).
8. $\langle 4 \rangle = \{0,4,8\}$ has 3 elements, so the index is $12/3 = 4$ (Definition 10.10; Theorem 10.7).
9. $(1,2,3)$ is even and $(4,5)$ is odd, so the product is odd: $(1,2,3)$ is a product of two transpositions, $(4,5)$ is one transposition (Definition 8.22).

Section C.

10. $a \sim_L b$ if and only if $a^{-1}b \in H$ is reflexive, symmetric and transitive (Theorem 10.1). Each class is aH, and each x lies in xH, so the classes cover G and are disjoint.
11. $f(h) = ah$ is one-to-one by left cancellation and onto aH by definition (Theorem 10.6).
12. If $a^{-1}b \in K$ then $\phi(a) = \phi(b)$ because $\phi(a^{-1}b) = e$. If $\phi(a) = \phi(b)$ then $\phi(a^{-1}b) = \phi(a)^{-1}\phi(b) = e$, so $a^{-1}b \in K$ (Theorem 10.17).
13. $\operatorname{sgn}(\sigma\mu) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\mu)$: if σ and μ are products of p and q transpositions, $\sigma\mu$ is a product of $p+q$ transpositions (Section 8, sign of a permutation; Theorem 8.24).
14. A k-cycle is a product of $k-1$ transpositions (8.14), so it is even exactly when $k-1$ is even, that is when k is odd.

Section D.

15. Let $n = |G|$ and $m = |H|$. The cosets of H partition G and each has m elements (Theorem 10.6), so $n = rm$ where r is the number of cosets (Theorem 10.7). For $a \in G$, $\langle a \rangle$ has order m and $m \mid n$ (Theorem 10.9).
16. Write σ as cycles of lengths $\ell_1, \ldots, \ell_r$. A cycle of length ℓ is a product of $\ell - 1$ transpositions (8.14), so $\operatorname{sgn}(\sigma) = (-1)^{\sum(\ell_i - 1)}$, and $(-1)^{\ell_i - 1} = -1$ exactly when ℓ_i is even. So σ is odd exactly when the number of even-length cycles is odd (Theorems 8.19 and 8.24).

MODULE 3: SYLLABUS MAPPING

Topics the past papers do not reach, and the model questions that cover them

Topic	**Book section (8th ed.)**	**Model questions**
Disjoint cycles and transposition products	4.8 to 4.15; 8.14	5, 6
Orbits of a permutation	8.17 to 8.18 (Module II); cycles 4.11	3, 6
The alternating group and the sign homomorphism	8.19 to 8.25	4, 9, 13, 14, 16
Left cosets, partition, and cardinality	10.1 to 10.7	7, 10, 11, 15
Index	10.10 to 10.11	2, 8
Right cosets and kernel cosets	10.12 to 10.19	12
Parity of a permutation by cycle lengths	8.22 to 8.24; 4.11	14, 16